

Evaluation of Automatically Generated Process Performance Indicators

Thank you for taking the time to participate in this study.

Your input is very valuable for our research on process performance indicators (PPIs) and how they can be automatically derived from business process event logs.

In this study, you will be presented with sets of PPIs generated for two different business processes:

- Request for Reimbursement
- Incident Management

For each set of indicators, we ask you to evaluate their quality according to several criteria such as relevance, coverage, actionability, non-redundancy, and overall usefulness. Your responses will help us better understand how automatically generated indicators can support process monitoring and improvement. The questionnaire should take approximately 10 minutes to complete. All responses will be anonymous and will be used exclusively for research purposes. We sincerely appreciate your time and contribution to this study. By continuing with the questionnaire, you confirm your voluntary participation in this research.

* Indica que la pregunta es obligatoria

Participant background

1. Current country *

2. Current position or role *

3. Main areas of expertise *

4. Years of Professional Experience *

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- ☐ 0-1 years
- ☐ 2-5 years
- ☐ 6-10 years
- ☐ More than 10 years

5. To what extent are you experienced working with business process management or improvement initiatives? *

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	1	2	3	4	5	
Very	☐	☐	☐	☐	☐	Very high

6. To what extent are you experienced defining or using process performance indicators (PPIs or KPIs)? *

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	1	2	3	4	5	
Very	☐	☐	☐	☐	☐	Very high

7. To what extent are you familiar with incident management processes (e.g., IT support or service desk processes)? *

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1	2	3	4	5		
Very	☐	☐	☐	☐	☐	Very high

8. To what extent are you familiar with request for reimbursement processes (e.g., employee expense reimbursement)? *

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1	2	3	4	5		
Very	☐	☐	☐	☐	☐	Very high

Evaluation of PPI Sets

You will now see a list of PPIs generated for two given processes. Please evaluate the overall quality of each set of PPIs for each process using the following criteria.

Evaluation Criteria (5-point Likert) Scale:

- 1 – Not at all
- 2 – To a small extent
- 3 – To a moderate extent
- 4 – To a large extent
- 5 – To a very large extent

Relevance: The indicators are relevant for assessing the performance of the process. 1 – Not at all → 5 – To a very large extent.

Coverage: The indicators collectively cover the main aspects of the process performance. 1 – Not at all → 5 – To a very large extent.

Actionability: The indicators provide insights that could support decision-making or process improvement. 1 – Not at all → 5 – To a very large extent.

Non-Redundancy: The indicators are not unnecessarily repetitive or overlapping. 1 – Not at all → 5 – To a very large extent.

Usefulness: Overall, this set of indicators would be useful for monitoring and improving the process. 1 – Not at all → 5 – To a very large

Process 1 : Request for Reimbursement

In this request for reimbursement process, an employee submits a declaration, which may then go through approval steps involving supervisors and other organizational roles before payment is requested and handled. After submission by the employee, the request is sent for approval to the travel administration. If approved, the request is then forwarded to the budget owner and after that to the supervisor. If the budget owner and supervisor are the same person, then only one of these steps is taken. In some cases, the director also needs to approve the request. In all cases, a rejection leads to one of two outcomes. Either the employee resubmits the request, or the employee also rejects the request. If the approval flow has a positive result, the payment is requested and made.

PPI Set A

- Percentage of 'Declaration SUBMITTED by EMPLOYEE' out of all 'Declaration APPROVED by ADMINISTRATION'
- Number of 'Declaration SUBMITTED by EMPLOYEE' activities for cases where activity 'Request Payment' occurs
- Percentage of 'Declaration SUBMITTED by EMPLOYEE' out of all cases
- Percentage of 'Declaration SUBMITTED by EMPLOYEE' out of all 'Payment Handled'
- Percentage of 'Declaration SUBMITTED by EMPLOYEE' out of all 'Declaration FINAL_APPROVED by SUPERVISOR'
- Number of 'Declaration SUBMITTED by EMPLOYEE' activities for cases where activity 'Declaration REJECTED by ADMINISTRATION' does not occur
- The average of the duration between activity 'Declaration SUBMITTED by EMPLOYEE' and activity 'Declaration APPROVED by ADMINISTRATION'
- The average of the duration between activity 'Declaration SUBMITTED by EMPLOYEE' and activity 'Declaration FINAL_APPROVED by SUPERVISOR'
- The average of the duration between activity 'Declaration SUBMITTED by EMPLOYEE' and activity 'Payment Handled'
- The average of the duration between activity Declaration SUBMITTED by EMPLOYEE and the end of the case
- The sum of the duration between activity 'Declaration SUBMITTED by EMPLOYEE' and activity 'Request Payment'
- The minimum of the duration between activity 'Declaration SUBMITTED by EMPLOYEE' and activity 'Declaration APPROVED by ADMINISTRATION'

PPI Set B

- Median time from the first Declaration SUBMITTED by EMPLOYEE to the first Payment Handled, considering only cases where both activities occur at least once

- Median time from the first Declaration SUBMITTED by EMPLOYEE to the first Declaration FINAL_APPROVED by SUPERVISOR, considering only cases where both activities occur at least once
- Median time from the first Request Payment to the first Payment Handled, considering only cases where both activities occur at least once
- Average share of cases that: (1) contain at least one each of submitted, final-approved, request-payment, and payment-handled; (2) follow the sequence Submitted Final Approved, then Request Payment immediately after Final Approved, then Payment Handled immediately after Request Payment (with one match per Submitted); and (3) have zero Declaration REJECTED by events (any listed reject type)
- Percentage of cases with at least one declaration rejection of any listed type (employee, administration, supervisor, missing, pre-approver, or budget owner)
- Among cases with at least one Declaration SUBMITTED by EMPLOYEE, the percentage where that activity occurs more than once
- Among cases with at least one Request Payment, the percentage where Request Payment occurs exactly once
- Average share of cases where Request Payment occurs exactly once and Payment Handled occurs exactly once
- Average proportion of events performed by resource SYSTEM (SYSTEM event count divided by total event count)
- Average proportion of events with org:role = UNDEFINED (UNDEFINED event count divided by total event count)
- Median time from the first Declaration SUBMITTED by EMPLOYEE to the first Payment Handled for high-value cases (first case amount above the 75th percentile), considering only cases where both activities occur at least once
- Average share of cases with at least one Declaration APPROVED by BUDGET OWNER

9. **Relevance:** To what extent are the **PPIs in set A** relevant for assessing the performance of the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

10. **Relevance:** To what extent are the **PPIs in set B** relevant for assessing the performance of the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

11. **Coverage:** To what extent do the **PPIs in set A** collectively cover the main aspects of the process performance? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

12. **Coverage:** To what extent do the **PPIs in set B** collectively cover the main aspects of the process performance? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

13. **Actionability:** To what extent do the **PPIs in set A** provide insights that could support decision-making or process improvement? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

14. **Actionability:** To what extent do the **PPIs in set B** provide insights that could support decision-making or process improvement? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

15. **Non-Redundancy:** To what extent do the **PPIs in set A** avoid unnecessary repetition or overlap? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

16. **Non-Redundancy:** To what extent do the **PPIs in set B** avoid unnecessary repetition or overlap? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

17. **Usefulness:** To what extent would the **PPIs in set A** be useful for monitoring and improving the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

18. **Usefulness:** To what extent would the **PPIs in set B** be useful for monitoring and improving the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

19. **Understandability:** To what extent do you find the **PPIs in set A** understandable? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

20. **Understandability:** To what extent do you find the **PPIs in set B** understandable? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

21. Which set of indicators would you prefer to use as a starting point for process performance monitoring? *

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- ☐ Set A
- ☐ Set B
- ☐ No preference

22. Please briefly explain your choice in previous question

Process 2 : IT Incident Management

This event log records the execution of an IT incident management process, covering the lifecycle of incidents from queuing and assignment to resolution and closure. The log includes multiple states associated with incident handling, such as assignment, work in progress, waiting situations, resolution, and closure, which makes it suitable for analyzing operational performance aspects such as response, progression, and completion. The possible statuses are queued, accepted, completed and closes, while the possible substatuses are assigned, awaiting assignment, cancelled, closed, in progress and wait.

PPI Set C

- Average end-to-end process duration from the first process START timestamp to the last process END timestamp.
- Average time from the first Queued + Awaiting Assignment to the first Accepted + Assigned, considering only cases where both occur at least once.
- Share of cases where Queued + Awaiting Assignment occurs at least once.
- Average total time accumulated from occurrences of Accepted + Wait - User until the corresponding end points defined as True, for cases where Accepted + Wait - User occurs at least once.
- Share of cases where Accepted + Wait - Vendor occurs at least once.
- Share of cases where Accepted + Wait - Implementation occurs at least once.
- Share of cases that have at least one Completed + Resolved event or at least one Completed + Closed event.
- Share of cases where Completed + Cancelled occurs at least once.
- Share of cases with at least one occurrence of activity "Accepted + Assigned".
- Share of cases where the first recorded activity is Accepted + In Progress.
- Average number of times activity "Accepted + In Progress" occurs per case.
- Share of cases where Unmatched + Unmatched occurs at least once.

PPI Set D

- Frequency of activity 'Accepted + Assigned' for cases where activity 'Completed + Resolved' occurs
- Percentage of 'Accepted + Assigned' out of all 'Completed + Closed'
- Frequency of activity 'Accepted + Assigned' for cases where activity 'Queued + Awaiting Assignment' occurs
- Percentage of 'Accepted + Assigned' out of all 'Accepted + In Progress'
- Percentage of 'Accepted + Assigned' out of all 'Accepted + Wait - User'
- Percentage of 'Accepted + Assigned' out of all cases
- The average of the duration between activity 'Accepted + Assigned' and activity 'Completed + Resolved'
- The average of the duration between activity 'Accepted + In Progress' and activity 'Accepted + Assigned'
- The average of the duration between activity Accepted + Assigned and the end of the case

- The sum of the duration between activity Accepted + Assigned and the end of the case
- The average of the duration between activity 'Accepted + Assigned' and activity 'Completed + Closed'
- The sum of the duration between activity Accepted + Assigned and the end of the case

23. **Relevance:** To what extent are the **PPIs in set C** relevant for assessing the performance of the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

24. **Relevance:** To what extent are the **PPIs in set D** relevant for assessing the performance of the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

25. **Coverage:** To what extent do the **PPIs in set C** collectively cover the main aspects of the process performance? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

26. **Coverage:** To what extent do the **PPIs in set D** collectively cover the main aspects of the process performance? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

27. **Actionability:** To what extent do the **PPIs in set C** provide insights that could support decision-making or process improvement? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

28. **Actionability:** To what extent do the **PPIs in set D** provide insights that could support decision-making or process improvement? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

29. **Non-Redundancy:** To what extent do the **PPIs in set C** avoid unnecessary repetition or overlap? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

30. **Non-Redundancy:** To what extent do the **PPIs in set D** avoid unnecessary repetition or overlap? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

31. **Usefulness:** To what extent would the **PPIs in set C** be useful for monitoring and improving the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

32. **Usefulness:** To what extent would the **PPIs in set D** be useful for monitoring and improving the process? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

33. **Understandability:** To what extent do you find the **PPIs in set C** understandable? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

34. **Understandability:** To what extent do you find the **PPIs in set D** understandable? *

Marca solo un óvalo.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ To a very large extent

35. Which set of indicators would you prefer to use as a starting point for process performance monitoring? *

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- ☐ Set C
- ☐ Set D
- ☐ No preference

36. Please briefly explain your choice in previous question

37. Do you have any comments, suggestions, or observations regarding the PPIs presented in this questionnaire?

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